

5. Writing an academic scientific paper

Do research,

Never re-search !!!

Title

Author

Affiliation

Abstract

Keywords

1. Introduction

2. Method

3. Results and Discussion

4. Conclusions

Appendix

Acknowledgment

References

Title

The title should reflect the article's content clearly and precisely, and enable the reader to decide whether they want to read the article. The title is the main advertisement for the article – a great title will entice the audience to read on, whereas a poorly titled article may never reach its target audience. Keep the title simple, catchy and specific. Omit unnecessary words such as 'a study of', 'investigations of', 'observations on,' and avoid using abbreviations and jargon.

Authors and their Affiliation

The listing of authors should only include those who have made an intellectual contribution to the research, who will take public responsibility for the data and conclusions, and who have approved the final version of the manuscript.

Keyword list

A list of important words that reflect the research, in addition to those already used in the title. Choosing the right keywords will increase the likelihood of your article being found by other researchers, as keywords are used by abstracting and indexing services.

Abstract

The abstract should summarize the problem or study objective, the method, the results, and the conclusions of your research. The abstract enables you to elaborate on each major section of the article, usually in up to 200 words. The abstract should give sufficient detail that the reader can decide whether or not to read the whole article. Together, the title and the abstract should be able to stand on their own, as they are processed further by indexing services. While it's a good idea to make the abstract interesting, keep it accurate and relevant: Don't fall into the trap of trying to make your abstract promise more than the article can deliver.

Introduction

The introduction should be brief. It is meant to provide context and background, but should not be a history lesson. It should state clearly the problem being investigated, the background that puts the problem in context, and the reasons for conducting the research. You should summarize relevant research to provide context, and state the questions you are answering. Explain any findings of others that you are challenging or extending. Briefly and logically lead the reader to your hypothesis(es), research question(s), and general experimental design or method, if relevant.

Method *(Also called Materials and Methods or Experimental Methods)*

This section provides the readers with enough detail that they can replicate your research. Explain how you studied the problem, identify the procedures you followed, and structure these as logically as possible. If your methods are new, you will need to explain them in detail. If they have been published before, name the method and cite the previously published work; if you have modified the methods, refer to the original work and include your amendments. Identify the equipment and describe materials used and specify the source if there is variation in quality of materials. Include the frequency of observations, what types of data were recorded. Be precise in describing measurements and include strengths and weaknesses of measurement. Name any statistical tests used so that your numerical results can be validated. It is advisable to use the past tense, and avoid using the first person.

Results

This section should present your findings objectively, and explain what was found, largely in text. This is where you show how your new results contribute to the body of scientific knowledge, so it is important to be clear and set them out in a logical sequence. The sequence should be based on the tables and figures that best present your findings. Raw data are rarely included in a scientific article; instead the data are analyzed and presented in the form of figures, graphs, tables, and descriptions of observations. It is important to identify any significant trends clearly for the reader. Tables and figures must be numbered separately, and should be in the sequence that you refer to them in the text. Figures should have a brief description (a legend), providing the reader with sufficient information to know how the data were produced. It is important not to interpret your results - this should be done in the Discussion & Conclusions section.

Discussion and Conclusions

This section describes what your results mean, specifically in the context of what was already known about the subject of the investigation. You can present global and specific conclusions here, but be careful not to summarize your manuscript – this is what the abstract is for. You should link this section back to the introduction by referring to the question(s) or hypothesis(es), and indicate how the results relate to your expectations and to the literature cited. Do the results support or contradict previous theories? Most significantly, the discussion should explain how the research has moved the body of scientific knowledge forward. Suggest further experiments and indicate whether they are underway, and indicate uses and extensions if appropriate. Your conclusions should be supported by your results: It is important not to extend your conclusions beyond this, so avoid undue speculation and bold judgments about impact. This is also a good section to suggest practical applications of your results, and to outline what would be the next steps in your research.

Supplementary material (Appendix)

If you have additional content like raw data, program list, many photos, list of model, that you think would be useful for the reader, then you could include it as Appendix.

Acknowledgments

This section should be brief and include the names of individuals who have helped with your research, such as contributors, and suppliers who provided materials free of charge.

References

New research draws on knowledge obtained through previous research, and the source of this previously published work should be acknowledged. Any information not from your experiment and not 'common knowledge' should be recognized with a citation. The way citations are presented varies considerably, so to make sure your references are in the correct format. Any quoted text that appears in the article, long or short, should be within quotation marks and should include a reference. The references section at the end of the article includes all references cited in your article. This section is in contrast to a bibliography, common in books, where works read but not necessarily cited in the text are listed.

Language quality and editing

One important consideration is the quality of the language used: errors in grammar, spelling or punctuation will lead to rejection of the paper, and preventing the research from receiving the recognition it deserves.

1 THE EXPERIMENTAL RESEARCH REPORT

Title

Author

Affiliation

Abstract

Keywords

1. Introduction

2. Method

3. Results and Discussion

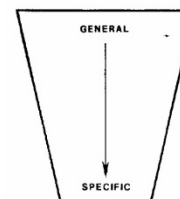
4. Conclusions

Appendix

Acknowledgment

References

2 THE INTRODUCTION: Establishing a Context



FIRST STAGE: General statement(s) about a field of research to provide the reader with a *setting* for the problem to be reported

SECOND STAGE: More specific statements about the aspects of the problem *already studied* by other researchers

THIRD STAGE: Statement(s) that indicate *the need for more investigation*

FOURTH STAGE: Very specific statement(s) giving *the purpose/objectives* of the writer's study

FIFTH STAGE: Optional statement(s) that give a *value* or *justification* for carrying out the study

FIGURE 2.2 The five stages of the introduction.

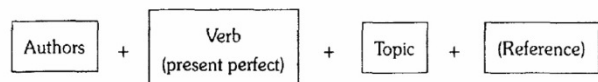
Guidelines for Marking Generic and Specific Noun Phrases

If you are having difficulty determining which, if any, article to use before a noun or noun phrase, ask yourself the following sequence of questions:

1. Is the noun meant in a *general* or a *specific* sense? If it is *specific*, use "the" before the noun. If it is *general*, ask yourself a follow-up question:
2. Is the noun *countable* or *uncountable*? If it is *countable*, use *a* or *an* (singular) or *-s* on the end (plural). If it is *uncountable*, use $\emptyset$ (no article or *-s* ending).

YOUR NOUN PHRASE		
1. GENERAL?	SPECIFIC?	
2. UNCOUNTABLE?	COUNTABLE?	<i>the</i> (add <i>-s</i> if plural)
$\emptyset$	<i>a/an</i> (singular) <i>-s</i> (plural)	

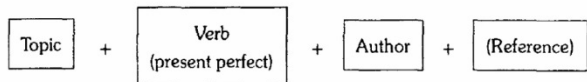
WEAK AUTHOR PROMINENT: Several Authors



Several
researchers

have studied

the relationship between
classroom adjustment and
mobility (Madsen, 1980;
Biggs, 1983; Randall,
1985).

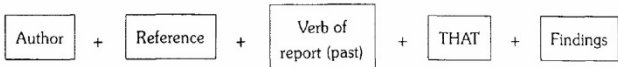


The
physiology
of annual
plants

has been studied

by several
authors (3, 7, 13).

AUTHOR PROMINENT: Findings from Specific Studies



Allington

(1983)

*found
showed
reported
noted
observed*

that

teachers
allocated
equal time
to all groups.

3 THE INTRODUCTION: Reviewing Previous Research

INFORMATION PROMINENT: Reporting Facts



Nutrient
resorption

is

a common
phenomenon
in forest trees

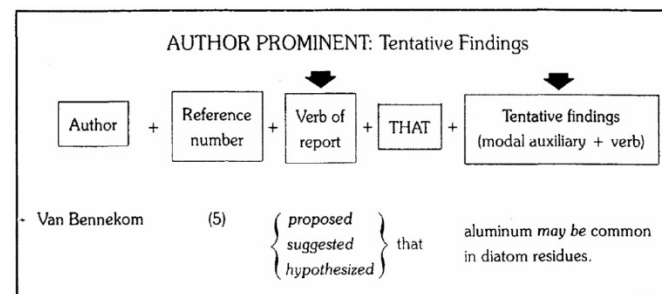
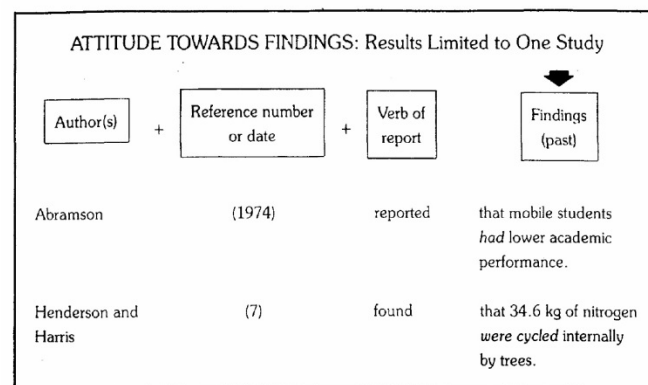
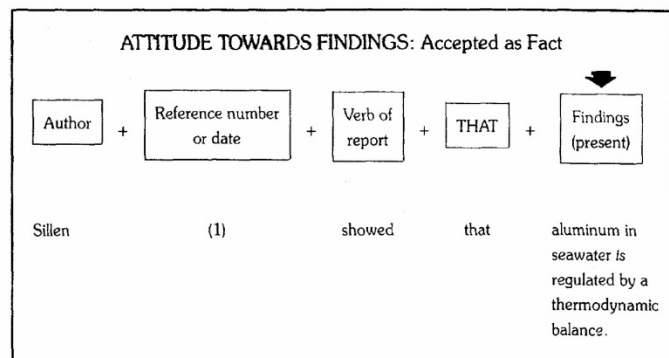
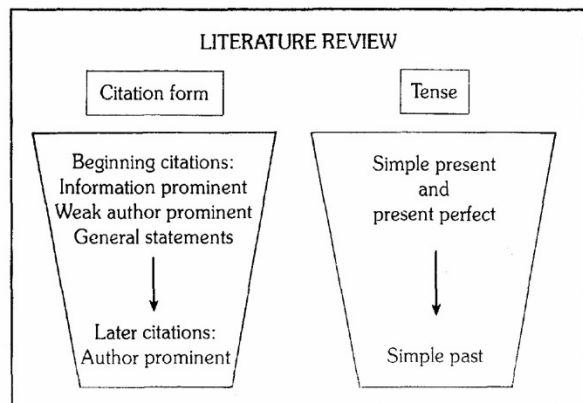
(Kramer, 1979).

The factors that
control the concen-
tration of aluminum
in seawater

are

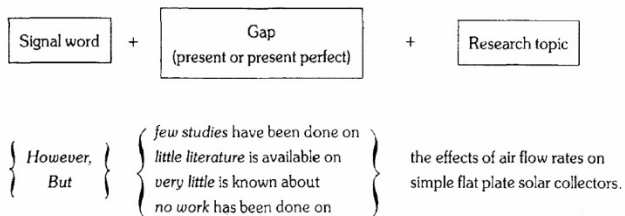
poorly known

(1).

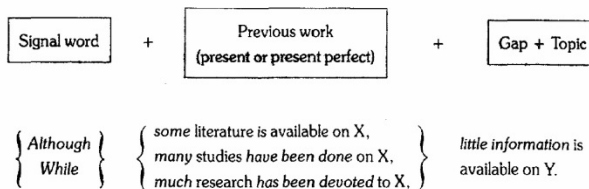


4 THE INTRODUCTION: Advancing to Present Research

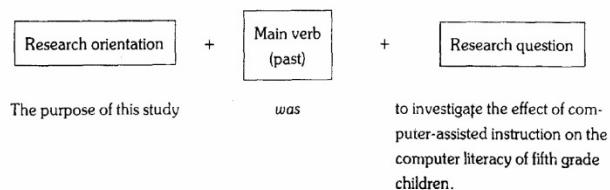
STAGE III: Signaling Missing Information



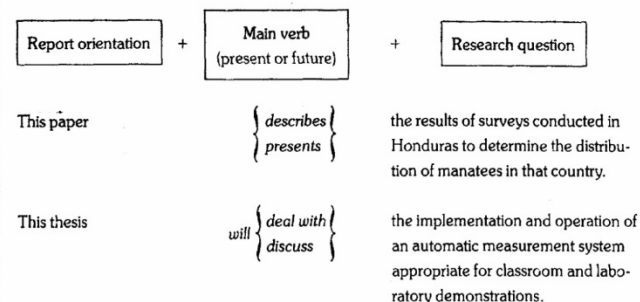
STAGE III: Signal Words (continued)



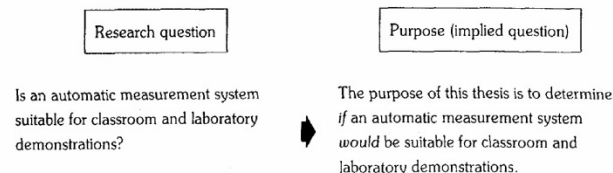
STAGE IV: Verb Tenses Research Orientation



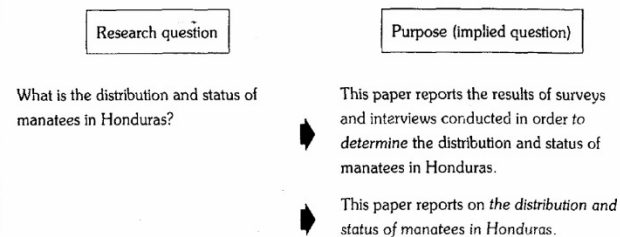
STAGE IV: Verb Tenses Report Orientation



STAGE IV: Implied Questions in the Statement of Purpose Yes or No Questions



STAGE IV: Implied Questions in the Statement of Purpose Information Questions

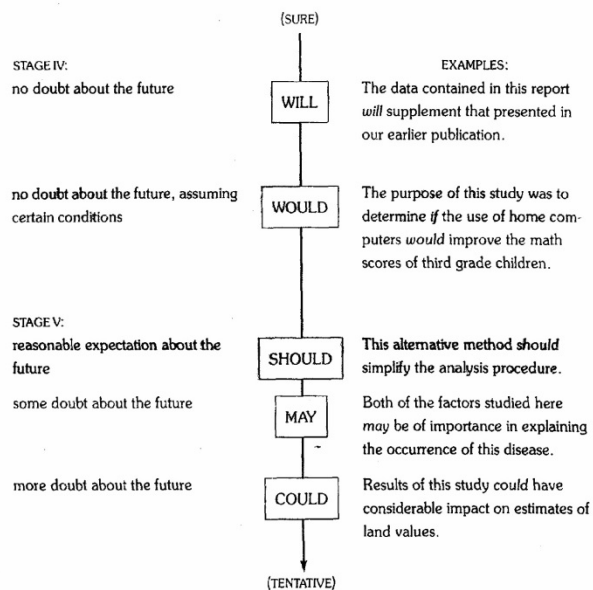


STAGE V: Statement of Value Using Modal Auxiliaries

Your research + Modal auxiliary + Value

The application of the strategies described here	{ may should }	increase the proportion of drug abusers who can be identified.
The system described here	could	serve as the basis for a study of automatic measurement systems in an instrumentation course.
This study	may	lead to a better understanding of phosphorus in natural systems.

MODAL AUXILIARIES: Degrees of Tentativeness



5 METHOD

PROCEDURAL STATEMENTS

ACTIVE VOICE

Agent + Main verb (active) + Object + Complement

We applied stress to the rubber segments in gradually increasing increments.

PASSIVE VOICE

Subject + Main verb (passive) + Agent + Complement

Stress was applied (by the investigators) to the rubber segments . . .

SHORTENING COMPOUND SENTENCES IN THE PASSIVE VOICE: Same Subjects

FULL FORM:

Subject₁ + be + Past participle₁ + CONJUNCTION + Subject₁ + be + Past participle₂

The data were collected and they were analyzed.

SHORT FORM:

The data were collected and analyzed.

SHORTENING COMPOUND SENTENCES IN THE PASSIVE VOICE: Different Subjects

FULL FORM:

Subject₁ + be + Past participle₁ + CONJUNCTION + Subject₂ + be + Past participle₂

The data were collected and correlations were calculated.

SHORT FORM:

The data were collected and correlations calculated.

6 MATERIALS

DESCRIBING SAMPLES: Past Tense Verbs

Sample	Main verb (past)	Description
The boys	were	between the ages of 7 and 13.
The men interviewed	were	primarily from St. Louis, Mo.
The subjects	were	18 Arabic-speaking students attending classes at the American University in Cairo.

DESCRIBING POPULATIONS: Present Tense Verbs

Population	Main verb (present)	Description
All students who apply for admission to the American University of Cairo	take	the Michigan Test of English Language Proficiency.
They	enter	the English Language Institute where they follow an intensive program of English language training.

DESCRIBING CONVENTIONAL MATERIAL: Present Tense Verbs

Conventional material	Main verb (present)	Description
The Auditory Test for Language Comprehension (Carrow, 1968)	permits	the assessment of oral language comprehension of English and Spanish.
A typical chemical reactor	includes	a helical, tube-in-tube heat exchanger.

1. The *passive voice* is usually used when a human agent (the experimenter) is manipulating the materials.

HUMAN AGENT INVOLVED: Passive Voice

- EXAMPLE A: The temperature inside the chamber *was increased* from 0° to 20 °C. (The researcher increased the temperature.)
- EXAMPLE B: Four thermocouples *were monitored hourly*. (A researcher monitored them.)

2. The *active voice* is usually used when no human is directly responsible for manipulating the materials—that is, when the materials operate “by themselves.”

NO HUMAN AGENT INVOLVED: Active Voice

- EXAMPLE C: A 200 hp generator *provided* power to the piezometers.
- EXAMPLE D: Control gauges *monitored* air pressure inside the chamber.

In examples C and D, the use of the active voice indicates that the experimenters were not directly involved in the functioning of the equipment.

3. The passive voice may be used to describe an action involving a nonhuman agent, but a *phrase* must be included to indicate the agent.

NO HUMAN AGENT INVOLVED: Passive Voice

- EXAMPLE E: Power *was supplied by 14 generators* with capacities ranging from 90 to 300 KW.

7 RESULTS

ELEMENT 1: LOCATING THE FIGURE

Present Tense

EXAMPLE: Results of the t-tests *are presented* in Table 1.

EXAMPLE: Table 4 *summarizes* the test results on precontaminated insulators.

ELEMENT 2: PRESENTING THE FINDINGS

Past Tense

EXAMPLE: As a group, divorced mothers *spent* over twice as much time in employment as married mothers (Figure 2).

EXAMPLE: The coefficient of correlation *was found* to be significant at the .001 level.

ELEMENT 3: COMMENTING ON THE RESULTS

Present Tense and Modal Auxiliaries

When the comment *compares* your results with the results of other studies, use the *present tense*.

EXAMPLE: This *is* consistent with earlier findings suggesting that personal characteristics *are* not related to attrition and teaching.

When the comment gives a *possible explanation* for the results, use a *modal auxiliary*.

EXAMPLE: These results *can* be explained by considering the voltage distribution on 230 kV insulators during freezing conditions.

When the comment *generalizes* from the results, use *may*.

EXAMPLE: Hyperactive children *may* be generally responsive to amphetamines.

ELEMENT 3: COMMENTING ON THE RESULTS

Tentative Verbs

EXAMPLE: It $\left\{ \begin{array}{l} \text{appears} \\ \text{seems} \\ \text{is likely} \end{array} \right\}$ that hyperactive children are generally responsive to amphetamines.

EXAMPLE: These results *suggest* that children who display learning problems are depending on only one cerebral hemisphere.

ELEMENT 2: COMPARISONS AMONG GROUPS

Group 1

Comparison

Group 2

The professional athletes

had *faster eye movements*

than our other subjects.

Quartz I reactors

had a *higher mass flow*

than Quartz II.

Superlative

Group 1

The *highest incidence* of Otitis Media

was found among Australian Indians.

ELEMENT 2: FLUCTUATION OF A VARIABLE OVER TIME

Variable

Verb

Phrase of variation

Time period

Prices

showed

a tendency to increase

over the three-year period.

The percentage of female students

tended

to decline

in the second half of the decade.

Variable

Verb of variation

Time period

The concentration of sulfur dioxide

*rose
fell
increased
decreased
dropped
remained constant
declined*

over the period studied.

ELEMENT 2: RELATIONSHIP BETWEEN TWO OR MORE VARIABLES

Variable X

Verb of correlation/association

Variable Y

Choice of location

was $\left\{ \begin{array}{l} \text{correlated with} \\ \text{negatively correlated with} \\ \text{associated with} \end{array} \right\}$

marital status.

Dry weight of top growth

was not $\left\{ \begin{array}{l} \text{highly} \\ \text{significantly} \\ \text{closely} \end{array} \right\}$ related to

total nitrogen.

8 DISCUSSION

COMPLEX SENTENCE STRUCTURE IN DISCUSSION STATEMENTS

Main clause (researcher's position)	+	THAT	+	Noun clause
We can conclude with certainty		that		both theories are able to explain significant amounts of variance.

VERB TENSES IN FIRST DISCUSSION ELEMENTS: Past, Present, and Modal Auxiliaries

Explaining the findings

EXAMPLE: It is possible that microbial activity *caused* some immobilization of labial soil phosphorous. (*restricted to study*)

EXAMPLE: It is possible that microbial activity *causes* some immobilization of labial soil phosphorous. (*general condition*)

Limiting the findings

EXAMPLE: Our sample *was* very small.

EXAMPLE: Other industries *may produce* different results.

When comparing your findings to those of other researchers, use the *present tense*.

VERB TENSES IN FIRST DISCUSSION ELEMENTS: Present Tense

Comparing findings

EXAMPLE: These results *are* in substantial agreement with those of Bates (2).

VERB TENSES IN LATER ELEMENTS: Present and Modal Auxiliaries/Tentative Verbs

Implications

EXAMPLE: It *appears* that squatter housing markets behave as economically rational entities.

Recommendations and applications

EXAMPLE: The approach outlined in this study *should be replicated* in other manufacturing plants.

EXAMPLE: We *recommend* that the approach outlined in this study be replicated in other manufacturing plants.

EXPRESSIONS FOR RESTATING THE HYPOTHESIS

Main clause
(position)

+

THAT

+

Noun clause
(information)

It was *anticipated*

The theory *led us to infer*

In line with this hypothesis, we *assumed*

The results *seem inconsistent* with our hypothesis

that

older workers in speed jobs would have poorer performance than younger workers.

EXPRESSIONS FOR EXPLAINING FINDINGS

Main clause
(position)

+

THAT

+

Noun clause
(information)

These results can be explained by *assuming*

that

skill increases with experience.

One reason *could be*

that

jobs in light manufacturing do not tax older workers to their limits.

It is *unlikely*

that

age of the subjects had much of an effect on attitude.

EXPRESSIONS FOR SUGGESTING IMPLICATIONS

Main clause
(position)

+

THAT

+

Noun clause
(information)

These findings

*suggest
imply
lend support to
the assumption
lead us to
believe
provide evidence*

that

frost affects the pan by breaking its massive structure.

9 ABSTRACT

ABSTRACT: Verb Tenses

B Background information (present tense)

EXAMPLE: One of the basic principles of communication *is* that the message should be understood by the intended audience.

P Principal activity (past tense/present perfect tense)

EXAMPLE: In this study the readability of tax booklets from nine states *was evaluated*.

EXAMPLE: Net energy analyses *have been carried out* for eight trajectories which convert energy source into heated domestic water.

M Methodology (past tense)

EXAMPLE: Children *performed* a 5-trial task.

R Results (past tense)

EXAMPLE: Older workers *surpassed* younger ones in both speed and skill jobs.

C Conclusions (present tense/tentative verbs/modal auxiliaries)

EXAMPLE: The results *suggest* that the presence of unique sets of industry factors *can be used* to explain variation in economic growth.

Conclusions

For a conclusion, you might detailed on the importance of the work or suggest the potential applications and extensions.

Appendix

Acknowledgment

References

Citations must be numbered consecutively in square brackets [1]. The list of references should be arranged in the order of citation in text.